

DISTRIBUTIONALLY ROBUST LEARNING FOR MULTI-SOURCE UNSUPERVISED DOMAIN ADAPTATION

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June 4, 2026



1 Introduction.

Our results and contribution.

Notations.

4 Bias Correction for the Matrix Γ

5 Theoretical Justification

6 Simulations.

Introduction

A fundamental assumption for the success of machine learning (ML) algorithms is that the training and test data sets share the same underlying distribution. However, when a domain shift occurs –that is, when the distribution of the test data differs from that of the training data –classical statistical learning algorithms often exhibit unreliable performance, even when they are carefully fine-tuned on the training data [43, 26, 29]. Developing methods that remain robust and achieve excellent prediction performance under domain shift is both critical and challenging.

Domain adaptation techniques have proven to be effective for improving generalization by transferring knowledge from one or multiple source domains to an unseen target domain [30, 16, 19, 48]. In contrast to traditional multi-source settings, where only data from multiple source domains are available, the multi-source unsupervised domain adaptation (MSDA) framework leverages not only multiple labeled source datasets but also unlabeled target data, as depicted in Figure 1. By incorporating the unlabeled target observations, MSDA may lead to better predictive models tailored for the target domain. This approach is particularly important in fields such as healthcare [50, 41] or finance [11], where the acquisition of target labels is often expensive or infeasible, and the target domain can differ substantially from any of the source domains. Meanwhile, compared to the semi-supervised setting, which considers more covariate data than outcome labels, this work focuses on the fully unsupervised setting for the target domain, where no target labels are available.

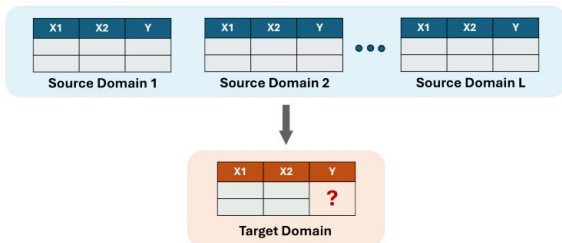


图 1: Illustration of the Multi-source Unsupervised Domain Adaptation (MSDA) framework, where source domains have labeled data and the target domain only has unlabeled data.

Formal Setup

We now introduce a formal setup of our focused MSDA regime.

Suppose that there are independent datasets collected from L source domains. For each source $1 \leq l \leq L$, the data $\{X_i^{(l)}, Y_i^{(l)}\}_{1 \leq i \leq n_l}$ are assumed to be i.i.d. samples generated according to:

$$X_i^{(l)} \sim \mathbf{P}_X^{(l)} \quad \text{and} \quad Y_i^{(l)} = f^{(l)}(X_i^{(l)}) + \varepsilon_i^{(l)}, \quad (1.1)$$

where $X_i^{(l)} \in \mathbb{R}^p$ and $Y_i^{(l)} \in \mathbb{R}$ represent the covariates and outcome respectively, and $f^{(l)}(x) = \mathbb{E}[Y_i^{(l)} | X_i^{(l)} = x]$ stands for the conditional outcome model for the l -th source domain. Here, both the covariate distribution $\mathbf{P}_X^{(l)}$ and the conditional outcome model $f^{(l)}(\cdot)$ are allowed to vary by source l .

Formal Setup (Continued)

Meanwhile, we denote the target distribution by $\mathbf{Q} = (\mathbf{Q}_X, \mathbf{Q}_{Y|X})$, which may differ from any of the source distributions in (1.1). Under the MSDA regime, although we have access to i.i.d. covariate observations $\{X_j^{\mathbf{Q}}\}_{1 \leq j \leq N}$ drawn from $\mathbf{Q}_X$, no corresponding outcome labels are available.

1 Introduction.

Our results and contribution.

Notations.

4 Bias Correction for the Matrix Γ

5 Theoretical Justification

⑥ Simulations.

We define an uncertainty set comprising target populations whose conditional outcome distributions are generated as mixtures from those of source populations. The proposed robust prediction model is then formulated to optimize the worst-case reward (about the explained variance) evaluated over this uncertainty set. In practical scenarios, domain experts may possess prior knowledge about the target population or seek to ensure robustness within a specific region around the observed mixture of source distributions. We incorporate such prior information into constructing the uncertainty set, which may improve the prediction accuracy of our proposed robust prediction model [23, 54].

Our primary contribution, stated in Theorem 2.1, demonstrates that our distributionally robust model, defined via a minimax optimization, can be identified (in the population sense) as a weighted average of the individual source models $\{f^{(l)}\}_{1 \leq l \leq L}$ in (1.1). The optimal aggregation weights are determined by solving a convex quadratic program, thereby simplifying the original minimax formulation and enabling easy implementation using standard quadratic programming solvers. Moreover, we introduce a novel bias correction step to obtain more accurate aggregation weights. We refer to our entire methodology as Distributionally Robust Learning (DRoL), which first fits individual source conditional mean models using any ML method of the user’s choice, and then aggregates these models into a robust prediction model specifically tailored for the target domain.

We leverage the weighted average formulation to establish the convergence rates of our proposed DRoL method in terms of both the objective function and the model parameters. We further demonstrate that the bias correction step improves these rates for certain regimes. As special but essential examples, we show that the estimation error for the model parameters achieves $n^{-1/2}$ in low-dimensional linear regression and $\sqrt{s \log p/n}$ in the high-dimensional setting, with s denoting the maximum sparsity level in the source domains.

As another contribution, we explore and establish identification results for distributionally robust models that use alternative loss functions, including standard squared error and regret loss [1]. Our analysis reveals that the choice of loss function is pivotal for the constructed robust models in the MSDA setting. Specifically, while models employing squared error loss are similarly expressed as weighted averages analogous to DRoL, their aggregation weights are sensitive to heterogeneous noise levels across source domains. Regret-based models effectively mitigate the impact of heterogeneous noise, yet become NP-hard to solve when general prior information about the target domain is incorporated. In contrast, our proposed robust model not only alleviates the impact of heterogeneous noise levels but also effectively accommodates the general prior information; see the detailed discussions in Section 2.3.

1. *Journal of Management Studies*, 1997, 34, 1, 1-14.

- For the MSDA setting, we introduce a theoretically justified framework for constructing distributionally robust models, where the identification result in Theorem 2.1 reduces the minimax optimization to a standard quadratic program.
- We explore various losses when constructing distributionally robust models, and demonstrate the unique advantages offered by the proposed reward function.
- We propose a bias-correction method to more accurately estimate the optimal aggregation weights, which are then used to construct the distributionally robust model. This method is compatible with various ML algorithms and achieves a faster convergence rate compared to direct plug-in approaches.

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Unsupervised Domain Adaptation. Inspired by the foundational work of [3], most unsupervised domain adaptation methods aim to align the distributions between source and target domains by minimizing a specific distance metric [22, 28, 44]. Another prominent approach employs adversarial training techniques to map source and target domains into a shared representation space, ensuring that a discriminator cannot distinguish between them [19, 59, 48]. These existing methods implicitly assume that the conditional outcome distributions given the covariates, or their learned representations, are similar across both source and target domains, which is particularly challenging to verify without labeled target data. In contrast, without requiring such a similarity condition, our work leverages the idea of DRO to develop a robust prediction model that maintains good performance as long as the target distribution resides within a predefined uncertainty set.

1000

Group DRO and agnostic federated learning. Existing approaches in Group DRO [47, 23, 68] and agnostic federated learning [37, 14] primarily aim to construct generalizable prediction models using data from source domains without using information from the target domain. In contrast, as shown in Figure 1, our work focuses on the MSDA setting, where, in addition to data from multiple sources, observations of target covariates are also available. We leverage this extra information to design the distributionally robust model tailored for this target domain. This difference in settings leads to distinct definitions of the uncertainty class and algorithmic designs, as detailed in Sections 2.1 and 3.4. Moreover, whereas Group DRO works typically focus on classification applications, our approach is specially designed for the regression task.

1000

Minimax Regret Optimization. Minimax regret models have been explored in various contexts to address noise sensitivity in distributional shifts. In single-source settings, [1] introduced a minimax regret framework to mitigate the impact of noise levels across source and target distributions. Recent extensions include multi-source linear regression [36] and conditional average treatment effect estimation in causal inference [70]. However, these works restrict the constraint set to simplexes or polyhedrons, ensuring computational tractability. In Section 2.3, we show that such regret-based approaches become NP-hard for general convex constraints in the MSDA setting, limiting their ability to incorporate flexible prior information about the target distribution. In contrast, our proposed reward-based models not only mitigate the impact of heterogeneous noise levels across source domains, but also integrate the prior in a computationally efficient manner; see Section 2.3 for the detailed discussions.

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Maximin Effect. Previous works [32, 8, 46, 21] studied the estimation and inference for the maximin effect, a particular type of robust prediction model using linear models. This paper extends the concept of the maximin effect to encompass ML algorithms, moving beyond the linear models. Moreover, our DRoL approach is designed to incorporate prior knowledge about the target distribution, resulting in a less conservative model.

1 Introduction.

Our results and contribution.

Notations.

4 Bias Correction for the Matrix Γ

5 Theoretical Justification

6 Simulations.

100

For a vector $x \in \mathbb{R}^p$, we define its ℓ_q norm as $\|x\|_q = (\sum_{i=1}^p |x_i|^q)^{1/q}$ for $q \geq 0$. For a numeric value a , we use notation a_p to represent a length p vector whose every entry is a . For a matrix A , we use $\|A\|_F$, $\|A\|_2^*$ and $\|A\|_\infty$ to denote its Frobenius norm, spectral norm, and element-wise maximum norm, respectively. We use I_p to denote the p -dimensional identity matrix. Let n_l and N represent the sample size for the l -th source domain and the target domain, respectively.

2

Group Distributionally Robust Prediction Models.

Identification for Distributionally Robust Prediction Models.

Exploration of Various Loss Functions for Distributionally Robust Models.

Regret-Based Robust Model

Extensions of the Robust Models

Distributionally Robust Prediction Models: Definition and Identification.

We consider the multi-source unsupervised domain adaptation (MSDA) setting, where we have labeled samples from L source domains and only unlabeled data from the target domain. For each source domain $l \in [L]$, let $\{X_i^{(l)}, Y_i^{(l)}\}_{i \in [n_l]}$ be i.i.d. samples drawn from the distribution $\mathbf{P}^{(l)} = (\mathbf{P}_X^{(l)}, \mathbf{P}_{Y|X}^{(l)})$. For the target domain, characterized by $\mathbf{Q} = (\mathbf{Q}_X, \mathbf{Q}_{Y|X})$, we observe only unlabeled covariate samples $\{X_j^{\mathbf{Q}}\}_{j \in [M]}$. Our goal is to build a prediction model that robustly generalizes to the target domain, accommodating both covariate shift (where $\mathbf{Q}_X$ may differ from any $\mathbf{P}_X^{(l)}$) and posterior drift (where both $\mathbf{Q}_X$ and $\mathbf{Q}_{Y|X}$ may differ from any $\mathbf{P}_X^{(l)}$ and $\mathbf{P}_{Y|X}^{(l)}$).

1 Introduction.

2 Distributionally Robust Prediction Models: Definition and Identification.

Group Distributionally Robust Prediction Models.

Identification for Distributionally Robust Prediction Models.

Exploration of Various Loss Functions for Distributionally Robust Models.

Regret-Based Robust Model

Extensions of the Robust Models

3 Algorithms: Distributionally Robust Learning

4 Bias Correction for the Matrix Γ

5 Theoretical Justification

Group Distributionally Robust Prediction Models.

In the absence of the labeled samples from the target domain, identifying its conditional outcome distribution $\mathbf{Q}_{Y|X}$ becomes particularly challenging, especially when we allow $\mathbf{Q}_{Y|X}$ to be different from the source domain's conditional outcome distributions. Instead of aiming at recovering $\mathbf{Q}_{Y|X}$, we propose a distributionally robust prediction model that ensures reliable predictive performance across a range of distributions, potentially encompassing the target $\mathbf{Q}$.

Group Distributionally Robust Prediction Models. (Continued)

In contrast to the unidentifiable $\mathbf{Q}_{Y|X}$, the target covariate distribution $\mathbf{Q}_X$ is empirically accessible via the available target covariate samples $\{X_j^{\mathbf{Q}}\}_{j \in [N]}$. Based on $\mathbf{Q}_X$, we define the uncertainty class of distributions as

$$\mathcal{C}(\mathbf{Q}_X) := \left\{ \mathbf{T} = (\mathbf{Q}_X, \mathbf{T}_{Y|X}) \mid \mathbf{T}_{Y|X} = \sum_{l=1}^L q_l \cdot \mathbf{P}_{Y|X}^{(l)} \quad \text{with} \quad q \in \Delta^L \right\}, \quad (2.1)$$

Uncertainty Set and Robust Prediction Model

- Define the uncertainty set for the target conditional distribution $Y|X$:

$$\mathcal{C}(\mathbf{Q}_X) := \left\{ \mathbf{T} = (\mathbf{Q}_X, \mathbf{T}_{Y|X}) \mid \mathbf{T}_{Y|X} = \sum_{l=1}^L q_l \cdot \mathbf{P}_{Y|X}^{(l)} \quad \text{with} \quad q \in \Delta^L \right\},$$

where $\Delta^L = \{q \in \mathbb{R}^L \mid \sum_{l=1}^L q_l = 1, \min\{q_l \geq 0\}\}$ denotes the $(L - 1)$ -dimensional simplex.

- Notably, if the true target conditional distribution $\mathbf{Q}_{Y|X}$ is indeed a mixture of $\{\mathbf{P}_{Y|X}^{(l)}\}_{l \in [L]}$ then this uncertainty set $\mathcal{C}(\mathbf{Q}_X)$ contains the actual target population $\mathbf{Q}$.

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- Unlike previous Group DRO works [47, 23, 68], which are designed for the regime without target covariate information and thus define uncertainty sets over the joint distribution of (X, Y) , our proposed uncertainty set is tailored to the target domain by fixing $\mathbf{Q}_X$ and focusing solely on the conditional distribution $Y|X$.
- In the no covariate shift scenario, the uncertainty sets defined by Group DRO and our approach coincide.
- However, in the presence of covariate shift, we utilize the available target covariate samples in the MSDA regime and focus solely on variations in the conditional distribution $Y|X$.

Reward Function

- Next, we introduce the reward function or the negative loss function, which serves as the basis for defining our distributionally robust prediction model.
- We will discuss the benefits of using our particular reward function over alternative loss functions in Section 2.3.
- Given a distribution T and a prediction model $f(\cdot)$, the reward function $\mathbf{R}_T(f)$ is defined as

$$\mathbf{R}_T(f) := \mathbb{E}_{(X,Y) \sim T}[Y^2 - (Y - f(X))^2],$$

where $\mathbb{E}_{(X,Y) \sim T}$ denotes the expectation taken with respect to the data (X, Y) following the distribution T .

Interpretation of the Reward Function

- The reward function $\mathbf{R}_T(f)$ evaluates the predictive performance of the model $f(\cdot)$ relative to the null model, evaluated on the distribution T .
- For the centered outcome variable (i.e., $\mathbb{E}_T[Y] = 0$), $\mathbf{R}_T(f)$ corresponds to the variance of the outcome explained by the model $f(\cdot)$.
- Therefore, a larger value of $\mathbf{R}_T(f)$ indicates better prediction performance of the model $f(\cdot)$.

Robust Prediction Model

- Building upon the uncertainty set $\mathcal{C}(\mathbf{Q}_X)$, we consider the worst-case reward of a prediction model f over this set, i.e. $\min_{\mathbf{T} \in \mathcal{C}(\mathbf{Q}_X)} \mathbf{R}_{\mathbf{T}}(f)$.
- We then introduce the robust prediction model f^* to maximize this worst-case reward, defined as

$$f^* := \arg \max_{f \in \mathcal{F}} \min_{\mathbf{T} \in \mathcal{C}(\mathbf{Q}_X)} \mathbf{R}_{\mathbf{T}}(f),$$

where $\mathcal{F}$ denotes a pre-specified function class.

Minimax Formulation

- Using the loss function terminology, we rewrite (2.3) into an equivalent minimax optimization as follows:

$$f^* := \arg \min_{f \in \mathcal{F}} \max_{\mathbf{T} \in \mathcal{C}(\mathbf{Q}_X)} \mathbb{E}_{\mathbf{T}}[\ell(X, Y; f)] \quad \text{with} \quad \ell(x, y; f) = (y - f(x))^2 - y^2.$$

- Since f^* optimizes the worst-case reward over the entire convex hull of $\{\mathbf{P}_{Y|X}^{(l)}\}_{l \in [L]}$, this can be conservative.

Incorporating Prior Knowledge

- A common strategy to mitigate this conservatism is to incorporate prior knowledge about the target distribution, thereby refining the uncertainty set [23, 37, 68, 54].
- For instance, prior expertise suggests that the target's $\mathbf{Q}_{Y|X}$ is a mixture of the sources' $\{\mathbf{P}_{Y|X}^{(l)}\}_{l \in [L]}$ with mixture weights close to a pre-specified weight vector $q^{\text{prior}} \in \Delta^L$.
- In this case, we can restrict the target mixture weights to lie within the refined region $\mathcal{H} = \{q \in \Delta^L \mid \|q - q^{\text{prior}}\|_2 \leq \rho\}$, where $\rho > 0$ is a user-specified parameter controlling the size of $\mathcal{H}$.

Refined Uncertainty Set

- Let $\mathcal{H} \subseteq \Delta^L$ represent the prior knowledge regarding the sourcing mixture for the target domain.
- We define the refined uncertainty set as

$$\mathcal{C}(\mathbf{Q}_X, \mathcal{H}) := \left\{ \mathbf{T} = (\mathbf{Q}_X, \mathbf{T}_{Y|X}) | \mathbf{T}_{Y|X} = \sum_{l=1}^L q_l \cdot \mathbf{P}_{Y|X}^{(l)} \quad \text{with} \quad q \in \mathcal{H} \right\}$$

- This uncertainty set $\mathcal{C}(\mathbf{Q}_X, \mathcal{H})$ is a subset of $\mathcal{C}(\mathbf{Q}_X)$ defined in (2.1), thereby narrowing the range of plausible distributions by leveraging the prior information encapsulated in $\mathcal{H}$.

Refined Robust Prediction Model

- With the refined $\mathcal{C}(\mathbf{Q}_X, \mathcal{H})$, we define the corresponding robust prediction model $f_{\mathcal{H}}^*$ as:

$$f_{\mathcal{H}}^* := \arg \max_{f \in \mathcal{F}} \min_{\mathbf{T} \in \mathcal{C}(\mathbf{Q}_X, \mathcal{H})} \mathbf{R}_{\mathbf{T}}(f).$$

- The specification of $\mathcal{C}(\mathbf{Q}_X, \mathcal{H})$ involves balancing predictiveness against the robustness of $f_{\mathcal{H}}^*$.
- If $\mathcal{C}(\mathbf{Q}_X, \mathcal{H})$ contains the target distribution $\mathbf{Q}$, this smaller uncertainty set facilitates a more accurate prediction model.

Balancing Predictiveness and Robustness

- In contrast, a larger uncertainty set $\mathcal{C}(\mathbf{Q}_X)$, which does not incorporate prior knowledge $\mathcal{H}$, is more likely to contain $\mathbf{Q}$ and thus produce a more robust but potentially conservative prediction model.
- We recommend specifying a reasonable $\mathcal{H}$ based on domain expertise, with $\mathcal{H} = \Delta^L$ serving as a default where prior knowledge is limited.
- We have explored how different specifications of $\mathcal{H}$ affect the model predictiveness and robustness in the numerical studies in Sections 5.2 and 6.
- We demonstrate in Sections 5.2 and 6 that integrating such prior information can significantly mitigate the conservativeness and improve the prediction accuracy of the robust prediction model.

Connection with Minimax Fairness

- We conclude this subsection with a remark on the connection with minimax fairness.
- In scenarios without covariate shift (i.e. $\mathbf{Q}_X = \mathbf{P}_X^{(l)}$ for all l), our robust model (2.3) aligns with the minimax fairness framework [31, 15]; see the details in Section A.2 of Supplements [62].
- In the covariate shift setting, our proposal serves as a distributionally robust model in the MSDA regime, but it remains unclear whether it can be directly related to minimax fairness.

1 Introduction.

② Distributionally Robust Prediction Models: Definition and Identification.

Group Distributionally Robust Prediction Models.

Identification for Distributionally Robust Prediction Models.

Exploration of Various Loss Functions for Distributionally Robust Models.

Regret-Based Robust Model

Extensions of the Robust Models

4 Bias Correction for the Matrix Γ

5 Theoretical Justification

1. *Journal of Management Studies*, 1997, 34, 1, 1-14.

THEOREM 2.1

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Suppose that the function class $\mathcal{F}$ is convex with $f^{(l)} \in \mathcal{F}$ for all $l \in [L]$ and $\mathcal{H}$ is a convex subset of Δ^L , then $f_{\mathcal{H}}^*$ defined in (2.5) is identified as:

$$f_{\mathcal{H}}^* = \sum_{l=1}^L q_l^* \cdot f^{(l)} \quad \text{with} \quad q^* = \arg \min_{q \in \mathcal{H}} q^\top \Gamma q,$$

where $\Gamma_{k,l} = \mathbb{E}_{\mathbf{Q}_X}[f^{(k)}(X)f^{(l)}(X)]$ for $k, l \in [L]$.

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Figure 2: Geometric Interpretation

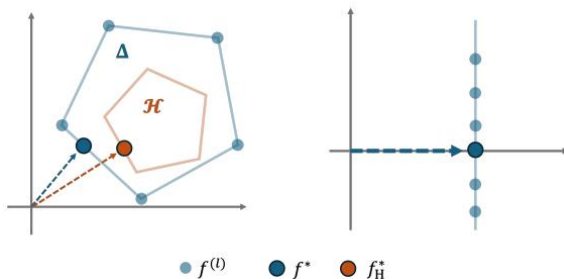


图 2: f^* (red point) and $f_{\mathcal{H}}^*$ (blue point) for $p = 2, L = 5$, and additive models. The left panel: f^* is the point closest to the origin in the convex hull of $\{f^{(l)}\}_{l \in [L]}$, and $f_{\mathcal{H}}^*$ is the point in the $\mathcal{H}$ -constrained set having the smallest distance to the origin; The right panel: consider the setting with shared first component $f_1^{(1)} = f_1^{(2)} = \dots = f_1^{(L)} = f_1$ and the second component $f_2^{(l)}$ being scattered around 0; the distributional robust prediction model $f^*(x) = f_1(x_1)$ (blue point) retains only the shared component and shrinks the sign heterogeneous component to zero.

1. **Introduction**

1. *Journal of the American Medical Association*, 1997; 277: 1039-1043.

- $$\min_{f \in \mathcal{F}} \max_{\mathbf{T} \in \mathcal{C}(\mathbf{Q}_X)} \mathbb{E}_{\mathbf{T}}[\ell(X, Y; f)] = \min_{f \in \mathcal{F}} \max_{1 \leq l \leq L} \mathbb{E}_{(\mathbf{Q}_X, \mathbf{P}_{Y|X}^{(l)})} [\ell(X, Y; f)],$$

1 Introduction.

② Distributionally Robust Prediction Models: Definition and Identification.

Group Distributionally Robust Prediction Models.

Identification for Distributionally Robust Prediction Models.

Exploration of Various Loss Functions for Distributionally Robust Models.

Regret-Based Robust Model

Extensions of the Robust Models

4 Bias Correction for the Matrix Γ

⑤ Theoretical Justification

- The choice of the loss function is crucial for constructing distributionally robust models in the MSDA setting.
- We establish identification results for robust models that use different loss functions, including the standard squared error and regret loss [?].
- We discuss the advantages of using the reward function that we have mainly adopted in the present paper.

Squared Error Loss

We start with the squared error loss, which leads to the following robust prediction model:

$$f_{\mathcal{H}}^{\text{sq}} = \min_{f \in \mathcal{F}} \max_{\mathbf{T} \in \mathcal{C}(\mathbf{Q}_X, \mathcal{H})} \mathbb{E}_{(X,Y) \sim \mathbf{T}}[(Y - f(X))^2], \quad (2.7)$$

where $\mathcal{F}$ denotes a pre-specified function class, and $\mathcal{H} \subseteq \Delta^L$ encodes the prior information about the sourcing mixture in the target domain.

Noise Levels and Proposition 2.1

For each source domain, we define the noise level as

$[\sigma^{(l)}]^2 = \mathbb{E}[(\varepsilon_i^{(l)})^2 | X_i^{(l)}]$ with $\varepsilon_i^{(l)} = Y_i^{(l)} - f^{(l)}(X_i^{(l)})$. We further define the vector $\boldsymbol{\sigma}^2 = ((\bar{\sigma}^{(1)})^2, \dots, (\bar{\sigma}^{(L)})^2)^\top$.

PROPOSITION 2.1

Suppose that the function class $\mathcal{F}$ is convex with $f^{(l)} \in \mathcal{F}$ for all $l \in [L]$ and H is a convex subset of Δ^L , then $f_{\mathcal{H}}^{\text{sq}}$ defined in (2.7) is identified as:

$$f_{\mathcal{H}}^{\text{sq}} = \sum_{l=1}^L q_l^{\text{sq}} \cdot f^{(l)} \quad \text{with} \quad q^{\text{sq}} = \arg \min_{q \in \mathcal{H}} \{q^\top \Gamma q - q^\top (\gamma + \boldsymbol{\sigma}^2)\}, \quad (2.8)$$

where $\Gamma_{k,l} = \mathbb{E}_{\mathbf{Q}_x} \mathbb{E}[f^{(k)}(X)f^{(l)}(X)]$ for $k, l \in [L]$, and $\gamma_l = \Gamma_{l,l}$ for $l \in [L]$.

Implications and Corollary 2.1

- This proposition indicates that $f_{\mathcal{H}}^{\text{sq}}$ is also characterized by aggregating the source models $\{f^{(l)}\}_{l \in [L]}$.
- The key distinction from the identification Theorem 2.1 is that the noise levels σ^2 across different source domains affect the aggregation weights of $f_{\mathcal{H}}^{\text{sq}}$.
- We demonstrate the disadvantage of such aggregation in the following corollary by considering a simpler regime with $L = 2$ and $\mathcal{H} = \Delta^2$.

COROLLARY 2.1

Suppose $L = 2$, and $\mathcal{H} = \Delta^2$. Then f^{sq} defined in (2.7) is identified as:

$$f^{\text{sq}} = q_1 f^{(1)} + (1 - q_1) f^{(2)} \quad \text{with} \quad q_1 = 0 \vee \left(\frac{1}{2} + \frac{\sigma_1^2 - \sigma_2^2}{\mathbb{E}_{\mathbf{Q}}[(f^{(1)}(X) - f^{(2)}(X))^2]} \right) \wedge$$

Consequently, when $\sigma_1^2 \gg \sigma_2^2$, $f^{\text{sq}} = f^{(1)}$; when $\sigma_1^2 = \sigma_2^2$, $f^{\text{sq}} = \frac{1}{2} f^{(1)} + \frac{1}{2} f^{(2)}$.

Regret Loss

The above corollary shows that f^{sq} prioritizes the group with the higher noise level, especially when one group exhibits a dominant noise over the other. When both groups have the same noise level, then f^{sq} becomes a simple average.

We now introduce the distributionally robust model based on the regret function in the MSDA setting, where the following regret function has been introduced in the work [?]

$$\text{Regret}_{\mathbf{T}}(f) = \mathbb{E}_{\mathbf{T}}[Y - f(X)]^2 - \inf_{f' \in \mathcal{F}} \mathbb{E}_{\mathbf{T}}[Y - f'(X)]^2.$$

The regret $\text{Regret}_{\mathbf{T}}(f)$ measures the excess risk of the model f compared to the optimal model for the distribution $\mathbf{T}$.

Regret-Based Robust Model and Proposition 2.2

Given the set $\mathcal{H} \subseteq \Delta^L$ and the function class $\mathcal{F}$, we define

$$f_{\mathcal{H}}^{\text{reg}} := \arg \min_{f \in \mathcal{F}} \max_{\mathbf{T} \in \mathcal{C}(\mathbf{Q}_x, \mathcal{H})} \text{Regret}_{\mathbf{T}}(f). \quad (2.9)$$

PROPOSITION 2.2

Suppose that the function class $\mathcal{F}$ is convex with $f^{(l)} \in \mathcal{F}$ for all $l \in [L]$, and $\mathcal{H}$ is a convex subset of Δ^L . Then $f_{\mathcal{H}}^{\text{reg}}$ is equivalent to:

$$f_{\mathcal{H}}^{\text{reg}} = \arg \min_{f \in \mathcal{F}} \left\{ r : \max_{q \in \mathcal{H}} \mathbb{E}_{\mathbf{Q}_X} \left[\left(\sum_{l=1}^L q_l f^{(l)}(X) - f(X) \right)^2 \right] \leq r \right\}. \quad (2.10)$$

Interpretation and Computational Challenge

- Therefore, $f_{\mathcal{H}}^{\text{reg}}$ corresponds to the center of the smallest circle enclosing all functions of the form $\{\sum_{l=1}^L q_l f^{(l)}(\cdot) \mid q \in \mathcal{H}\}$, known as the Chebyshev center.
- For general convex subsets $\mathcal{H} \subseteq \Delta^L$, solving (2.10) is NP-hard.
- This proposition reveals that identifying $f_{\mathcal{H}}^{\text{reg}}$ reduces to solving the Chebyshev center problem.
- For an arbitrary convex set $\mathcal{H} \subseteq \Delta^L$, computing the Chebyshev center is computationally intractable [?], except in special cases such as polyhedral [?] or finite sets [?].
- For example, when $\mathcal{H}$ is defined as the intersection of multiple ellipsoids, solving the optimization problem in (2.10) becomes NP-hard [?].

11

- Therefore, the computational intractability limits the regret-based models' usefulness when integrating prior information via a convex set H .
- In contrast, our reward-based approach, introduced in Section 2.2 effectively incorporates prior information while maintaining computational efficiency.
- As demonstrated in Sections 5.2 and 6, this prior integration leads to the improved predictive performance of the proposed approach on the target domain.

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Notably, the proposed reward function is the only one that produces a robust prediction model both independent of noise levels and computationally feasible when incorporating general convex prior mixture information.

Special Cases for Regret-Based Model

We now move on to considering two special cases of H where the regret-based $f_{\mathcal{H}}^{\text{reg}}$ is computationally feasible and will provide further insights for this regret-based approach.

COROLLARY 2.2

We consider two specifications of H as follows:

- ① When $\mathcal{H} = \Delta^L$, f^{reg} is identified as:

2

Group Distributionally Robust Prediction Models.

Identification for Distributionally Robust Prediction Models.

Exploration of Various Loss Functions for Distributionally Robust Models.

Regret-Based Robust Model

Extensions of the Robust Models

Regret-Based Robust Model

Definition

The regret-based robust model is defined as:

$$f^{\text{reg}} = \sum_{l=1}^L q_l^{\text{reg}} \cdot f^{(l)} \quad \text{with} \quad q^{\text{reg}} = \arg \min_{q \in \Delta^L} \{q^\top \Gamma q - q^\top \gamma\}, \quad (2.11)$$

where $\Gamma_{k,l} = \mathbb{E}_{\mathbf{Q}_X}[f^{(k)}(X)f^{(l)}(X)]$ for $k, l \in [L]$, and $\gamma_l = \Gamma_{l,l}$ for $l \in [L]$.

Regret-Based Robust Model (Cont.)

Constrained Formulation

When $\mathcal{H}$ is defined as:

$$\mathcal{H} := \left\{ q \in \Delta^L \mid \mathbb{E}_{\mathbf{Q}_X} \left[\sum_{l=1}^L (q_l - q_l^{\text{prior}}) f^{(l)}(X) \right]^2 \leq \rho \right\}, \quad (2.12)$$

where $q^{\text{prior}} \in \Delta^L$ denotes a prior weight vector and $\rho \geq 0$ controls the size of $\mathcal{H}$.

- Suppose q^{prior} lies strictly within Δ^L and ρ is small enough such that $\mathcal{H}$ does not intersect the boundary of Δ^L .
- Then $f_{\mathcal{H}}^{\text{reg}} = \sum_{l \in [L]} q_l^{\text{prior}} f^{(l)}$, which is independent of the value of ρ .

Interpretation and Comparison

- Result (a) is closely related to the works of [?] and [?], who establish similar results for linear models and conditional average treatment effects in causal inference, respectively.
- However, these works restrict the constraint set to simplexes or polyhedrons, whereas Proposition 2.2 shows that regret-based models become NP-hard for more general convex constraints.
- For $\mathcal{H} = \Delta^L$, we compare its geometric interpretation with our proposed reward-based model f^* and the squared-error based model f^q .

Geometric Interpretation for $L = 3$

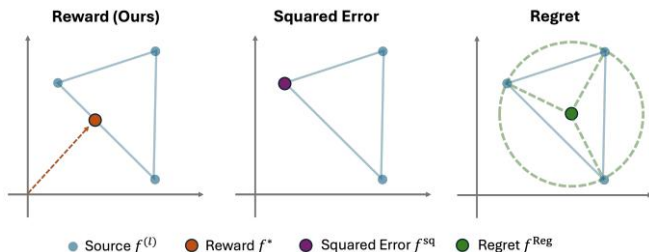


图 3: Illustration of robust prediction models utilizing reward (ours), squared error, and regret, for $L = 3$ and $\mathcal{H} = \Delta^3$.

Geometric Interpretation (Cont.)

- **Left panel:** f^* is the point closest to the origin within the convex hull of $\{f^{(l)}\}_{l \in [L]}$.
- **Middle panel:** f^{sq} corresponds to the source model with the largest noise level when this noise is substantially higher than that in other sources.
- **Right panel:** f^{reg} is the center of the smallest circle enclosing all individual source models.

Method	Source $f^{(l)}$	Regret f^{Reg}
Reward (Ours)	0	0
Regret	0	0

Discussion on Prior Weights

- For Result (b) in Corollary 2.2, $\mathcal{H}$ is specified as (2.12), encoding prior information that the conditional mean of the future target lies close to $\sum_{l \in [L]} q_l^{\text{prior}} f^{(l)}(X)$.
- Corollary 2.2 reveals that $f_{\mathcal{H}}^{\text{reg}}$ reduces exactly to $\sum_{l \in [L]} q_l^{\text{prior}} f^{(l)}(X)$, independent of the radius ρ .
- This implies the regret-based model $f_{\mathcal{H}}^{\text{reg}}$ is determined solely by the prior weights.
- In practice, q^{prior} may not be accurate, and one would expect the diameter ρ to play a role in reflecting the uncertainty of this prior.

2

Group Distributionally Robust Prediction Models.

Identification for Distributionally Robust Prediction Models.

Exploration of Various Loss Functions for Distributionally Robust Models.

Regret-Based Robust Model

Extensions of the Robust Models

1. *Journal of Management Studies*, 1997, 34, 1, 1-15.

$$\mathcal{C}'(\mathbf{Q}_X, \mathcal{H}) := \left\{ \mathbf{T} = (\mathbf{Q}_X, \mathbf{T}_{Y|X}) \mid \mathbb{E}_{\mathbf{T}}[Y|X] = \sum_{l=1}^L q_l \cdot \mathbb{E}_{\mathbf{P}^{(l)}}[Y|X], \quad q \in \mathcal{H} \right\}. \quad (2.13)$$

- $\mathcal{C}'(\mathbf{Q}_X, \mathcal{H})$ is broader than $\mathcal{C}(\mathbf{Q}_X, \mathcal{H})$, with $\mathcal{C}(\mathbf{Q}_X, \mathcal{H}) \subseteq \mathcal{C}'(\mathbf{Q}_X, \mathcal{H})$.
- It only requires a mixture of the conditional means rather than the entire conditional distributions.

Extended Robust Prediction Model

Definition

With a slight abuse of notation, we denote the robust prediction model based on $\mathcal{C}'(\mathbf{Q}_X, \mathcal{H})$ also by $f_{\mathcal{H}}^*$:

$$f_{\mathcal{H}}^* = \arg \min_{f \in \mathcal{F}} \max_{\mathbf{T} \in \mathcal{C}'(\mathbf{Q}_X, \mathcal{H})} \mathbf{R}_{\mathbf{T}}(f).$$

- $f_{\mathcal{H}}^*$ admits the same identification result as Theorem 2.1; see Supplement's Section A.1 [?].
- Furthermore, the reward function $\mathbf{R}_{\mathbf{T}}(f)$ can be extended to compare $f(\cdot)$ with any deterministic benchmark function.
- The identification theorem for this more general reward function is provided in Section A.1 of Supplements [?].

- ## Bias Correction: Main Idea

3. Algorithms: Distributionally Robust Learning

We leverage Theorem 2.1 to devise an algorithm to estimate $f_{\mathcal{H}}^*$.

- 1 For each source domain $l \in [L]$, use labeled data $\{X_i^{(l)}, Y_i^{(l)}\}_{i \in [n_l]}$ to fit the individual source model $f^{(l)}$ using a preferred ML method (e.g., random forests, boosting, neural networks).
- 2 The fitted model is denoted by $\hat{f}^{(l)}$.
- 3 Construct the plug-in estimator of $f_{\mathcal{H}}^*$ via aggregation:

$$\tilde{f}_{\mathcal{H}} = \sum_{l=1}^L \tilde{q}_l \cdot \hat{f}^{(l)} \quad \text{with} \quad \tilde{q} = \arg \min_{q \in \mathcal{H}} q^{\top} \tilde{\Gamma} q, \quad (3.1)$$

where $\tilde{\Gamma} \in \mathbb{R}^{L \times L}$ is the data-dependent estimator of Γ with $\tilde{\Gamma}_{k,l} = \frac{1}{N} \sum_{j=1}^N \hat{f}^{(k)}(X_j^{\mathbf{Q}}) \hat{f}^{(l)}(X_j^{\mathbf{Q}})$ for $k, l \in [L]$.

1

(x)

$$\hat{f}_{\mathcal{H}} = \sum_{l=1}^L \hat{q}_l \cdot \hat{f}^{(l)} \quad \text{with} \quad \hat{q} = \arg \min_{q \in \mathcal{H}} q^{\top} \hat{\Gamma} q. \quad (3.2)$$

- We explain the main idea behind this bias-corrected approach in Section 3.1.
- Detailed construction is provided in Sections 3.2 (for the no covariate shift setting) and 3.3 (for the general covariate shift setting).

- ### 3 Algorithms: Distributionally Robust Learning

$$\begin{aligned} \tilde{\Gamma}_{k,l} - \Gamma_{k,l} &= \left\{ \frac{1}{N} \sum_{j=1}^N \hat{f}^{(k)}(X_j^{\mathbf{Q}}) \hat{f}^{(l)}(X_j^{\mathbf{Q}}) - \mathbb{E}_{\mathbf{Q}_X} [\hat{f}^{(k)}(X) \hat{f}^{(l)}(X)] \right\} \\ &+ \left\{ \mathbb{E}_{\mathbf{Q}_X} [\hat{f}^{(k)}(X) \hat{f}^{(l)}(X)] - \mathbb{E}_{\mathbf{Q}_X} [f^{(k)}(X) f^{(l)}(X)] \right\}. \end{aligned}$$

The first term on the right-hand side represents the finite-sample variability, vanishing at the parametric rate, whereas the second term captures the estimation error of the product $(\widehat{f}^{(k)}\widehat{f}^{(l)})$.

1. *Journal of the American Medical Association*, 2000; 283: 2689-2695.

$$\underbrace{\mathbb{E}_{\mathbf{Q}_X} \left[\widehat{f}^{(k)}(X) \left(\widehat{f}^{(l)}(X) - f^{(l)}(X) \right) \right]}_{\text{Bias - kl}} + \underbrace{\mathbb{E}_{\mathbf{Q}_X} \left[\widehat{f}^{(l)}(X) \left(\widehat{f}^{(k)}(X) - f^{(k)}(X) \right) \right]}_{\text{Bias - lk}} - \underbrace{\mathbb{E}_{\mathbf{Q}_X} \left[\left(\widehat{f}^{(k)}(X) - f^{(k)}(X) \right) \left(\widehat{f}^{(l)}(X) - f^{(l)}(X) \right) \right]}_{\text{Higher order bias}}.$$

The “Higher-order bias” term typically diminishes at a faster rate and is negligible compared to the “Bias-kl” and “Bias-lk” terms, which are first-order error terms in either $\widehat{f}^{(k)} - f^{(k)}$ or $\widehat{f}^{(l)} - f^{(l)}$.

1 1 0 1

We employ sample-splitting to ensure independence. For each source $l \in [L]$, we randomly partition the data $\{X_i^{(l)}, Y_i^{(l)}\}_{i \in [n_l]}$ into two disjoint subsets $\mathcal{A}_l$ and B_l . We denote the fitted models from $\mathcal{A}_l$ and B_l by $\widehat{f}_A^{(l)}$ and $\widehat{f}_B^{(l)}$, respectively. We define:

$$(3.5) \quad \widehat{f}^{(l)}(X_i^{(l)}) = \begin{cases} \widehat{f}_{\mathcal{A}}^{(l)}(X_i^{(l)}) & \text{if } i \in \mathcal{B}_l \\ \widehat{f}_{\mathcal{B}}^{(l)}(X_i^{(l)}) & \text{if } i \in \mathcal{A}_l \end{cases}, \text{ and } \widehat{f}^{(l)}(X_j^{\mathbf{Q}}) = \frac{1}{2}\widehat{f}_{\mathcal{A}}^{(l)}(X_j^{\mathbf{Q}}) + \frac{1}{2}\widehat{f}_{\mathcal{B}}^{(l)}(X_j^{\mathbf{Q}})$$

This construction ensures $\hat{f}^{(l)}(\cdot)$ is independent of the sample it is applied to, which is essential for bias correction.

3.2. No Covariate Shift Setting

We first consider the setting without covariate shift, where $\mathbf{P}_X^{(l)} = \mathbf{Q}_X$ for all $l \in [L]$. In this case, each source data $\{X_i^{(l)}, Y_i^{(l)}\}_{i \in [n_l]}$ are i.i.d. samples from $(\mathbf{P}_X^{(l)}, \mathbf{P}_{Y|X}^{(l)}) = (\mathbf{Q}_X, \mathbf{P}_{Y|X}^{(l)})$. For $k, l \in [L]$, we correct the plug-in estimator $\tilde{\Gamma}_{k,l} = \frac{1}{N} \sum_{j=1}^N \hat{f}^{(k)}(X_j^{\mathbf{Q}}) \hat{f}^{(l)}(X_j^{\mathbf{Q}})$ by subtracting estimates of the dominant bias terms. The resulting bias-corrected estimator is $\hat{\Gamma}_{k,l} = \tilde{\Gamma}_{k,l} - \hat{\mathcal{D}}_{k,l} - \hat{\mathcal{D}}_{l,k}$, where

$$\hat{\mathcal{D}}_{k,l} = \frac{1}{n_l} \sum_{i=1}^{n_l} \hat{f}^{(k)}(X_i^{(l)}) \left(\hat{f}^{(l)}(X_i^{(l)}) - Y_i^{(l)} \right),$$

$$\hat{\mathcal{D}}_{l,k} = \frac{1}{n_l} \sum_{i=1}^{n_l} \hat{f}^{(l)}(X_i^{(k)}) \left(\hat{f}^{(k)}(X_i^{(k)}) - Y_i^{(k)} \right).$$

Here, $\widehat{D}_{k,l}, \widehat{D}_{l,k}$ estimate the "Bias-kl" and "Bias-lk" terms in (3.3), respectively.

In the covariate shift scenario, the target covariate distribution $\mathbf{Q}_X$ may differ from the source distributions $\{\mathbf{P}_X^{(l)}\}_{l \in [L]}$. We do not have direct access to data from $(\mathbf{Q}_X, \mathbf{P}_{Y|X}^{(l)})$ for bias correction. To address this, we adopt an importance weighting strategy. The key idea is to assign weights to the source data $\{X_i^{(l)}, Y_i^{(l)}\}_{i \in [n_l]}$ for each source $l \in [L]$, so that the reweighted source distribution resembles $(\mathbf{Q}_X, \mathbf{P}_{Y|X}^{(l)})$.

1. *Journal of the American Medical Association*, 2000; 283: 2686-2692.

For each source $l \in [L]$, suppose $\mathbf{Q}_X$ is absolutely continuous w.r.t. $\mathbf{P}_X^{(l)}$ and define the density ratio $\omega^{(l)}(x) = d\mathbf{Q}_X(x)/d\mathbf{P}_X^{(l)}(x)$. We estimate these density ratios $\{\omega^{(l)}\}_{l \in [L]}$ using sample-splitting. We construct $\hat{\omega}_A^{(l)}$ using data from split $\mathcal{A}_l$ and target covariates, and $\hat{\omega}_B^{(l)}$ using data from B_l and target covariates. We define the sample-split density ratio estimator:

$$\widehat{\omega}^{(l)}(X_i^{(l)}) = \begin{cases} \widehat{\omega}_{\mathcal{A}}^{(l)}(X_i^{(l)}) & \text{if } i \in \mathcal{B}_l \\ \widehat{\omega}_{\mathcal{B}}^{(l)}(X_i^{(l)}) & \text{if } i \in \mathcal{A}_l \end{cases}.$$

This ensures $\hat{\omega}^{(l)}(\cdot)$ is independent of the sample it is applied to.

Bias-Corrected Estimator under Covariate Shift

For $k, l \in [L]$, we construct the bias-corrected matrix estimator:

$$\hat{\Gamma}_{k,l} = \tilde{\Gamma}_{k,l} - \hat{\mathcal{D}}_{k,l} - \hat{\mathcal{D}}_{l,k},$$

where the plug-in estimator remains $\tilde{\Gamma}_{k,l} = \frac{1}{N} \sum_{j=1}^N \hat{f}^{(k)}(X_j^{\mathbf{Q}}) \hat{f}^{(l)}(X_j^{\mathbf{Q}})$, and the terms $\hat{\mathcal{D}}_{k,l}$, $\hat{\mathcal{D}}_{l,k}$ estimate "Bias-kl" and "Bias-lk" terms. In the covariate shift setting, these are:

$$\begin{aligned} \hat{\mathcal{D}}_{k,l} &= \frac{1}{n_l} \sum_{i=1}^{n_l} \hat{\omega}^{(l)}(X_i^{(l)}) \hat{f}^{(k)}(X_i^{(l)}) \left(\hat{f}^{(l)}(X_i^{(l)}) - Y_i^{(l)} \right), \\ \hat{\mathcal{D}}_{l,k} &= \frac{1}{n_k} \sum_{i=1}^{n_k} \hat{\omega}^{(k)}(X_i^{(k)}) \hat{f}^{(l)}(X_i^{(k)}) \left(\hat{f}^{(k)}(X_i^{(k)}) - Y_i^{(k)} \right). \end{aligned}$$

The source data are reweighted by the corresponding density ratio estimators. The no covariate shift case is a special instance with $\hat{w}^{(l)}(\cdot) \equiv \hat{w}^{(k)}(\cdot) \equiv 1$.

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1. *Journal of the American Medical Association*, 1997; 277: 1001-1005.

4. Theoretical Justification

In this section, we establish the convergence rates for the plug-in estimator $\hat{f}_{\mathcal{H}}$ defined in (3.1) and the bias-corrected estimator $\hat{f}_{\mathcal{H}}^{\text{bc}}$ defined in (3.2). We start with the following condition.

ASSUMPTION 1. The number of source domains L is finite. The matrix Γ defined in (2.6) is positive definite with $\lambda_{\min}(\Gamma) > 0$. There exists some positive constant $\sigma_{\varepsilon}^2 > 0$ such that

$$\max_{l \in [L]} \max_{i \in [n_l]} \mathbb{E}[(\varepsilon_i^{(l)})^2 | X_i^{(l)}] \leq \sigma_\varepsilon^2.$$

We assume the finite count of source domains to simplify the presentation, although our analysis can be extended to allow for growing L . The requirement of Γ being positive definite is ensured if the individual source models $\{f^{(l)}\}_{l \in [L]}$ are linearly independent. As for the noise level, we do not impose homogeneity across or within sources but only require its second moment to be bounded by some constant.

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For a prediction model $f(\cdot)$, we define its ℓ_q norm with respect to the target covariate distribution $\mathbf{Q}_X$ as $\|f\|_{\ell_q(\mathbf{Q})} := (\mathbb{E}_{\mathbf{Q}_X}[f(X)^q])^{1/q}$ for $q \geq 1$. We further define the scale of individual source models as

$$M := \max_{l \in [L]} \max \left\{ \|f^{(l)}\|_{\ell_2(\mathbf{Q})}, \|f^{(l)}\|_{\ell_4(\mathbf{Q})} \right\}.$$

Note that we do not assume M to be bounded; it is allowed to grow with the dimension p . For instance, in a high-dimensional linear setting where $f^{(l)}(x) = x^T \beta^{(l)}$ with $\beta^{(l)} \in \mathbb{R}^p$, we have $M \asymp \max_{l \in [L]} \|\beta^{(l)}\|_2$, which may increase with p .

4. Theoretical Justification (Cont.)

For each source domain $l \in [L]$, given the labeled data, we obtain the fitted individual source model $\hat{f}^{(l)}$ using the ML method of choice. Next, we assume that each $\hat{f}^{(l)}$ achieves a certain convergence rate in estimating the conditional mean model $f^{(l)}$; these rates will later be used to establish convergence results for our proposed DRoL estimators of the robust prediction model, as defined in (3.1) and (3.2). Let $n = \min_{l \in [L]} n_l$ denote the smallest sample size among L source domains. In the following assumption, we capture the convergence rate of the estimators $\{\hat{f}^{(l)}\}_{l \in [L]}$.

ASSUMPTION 2. With probability larger than $1 - \tau_n$ with $\tau_n \rightarrow 0$, there exists a positive sequence $\delta_n > 0$ such that $\max_{l \in [L]} \max\{\|\hat{f}^{(l)} - f^{(l)}\|_{\ell_2(\mathbf{Q})}, \|\hat{f}^{(l)} - f^{(l)}\|_{\ell_4(\mathbf{Q})}\} \leq \delta_n$.

Without further specification, we consider that each fitted individual model $\hat{f}^{(l)}$ is a consistent estimator of the true $f^{(l)}$ (i.e. $\delta_n \rightarrow 0$). Consistency has been established for various ML methods. For example, in high-dimensional sparse linear regression, estimators such as the Lasso or the Dantzig selector satisfy Assumption 2 with $\delta_n = C\sqrt{s_\beta \log p/n}$, where $s_\beta = \max_{l \in [L]} \|\beta^{(l)}\|_0$ denotes the largest sparsity level of $\beta^{(l)}$ for $l \in [L]$ [?, ?]. In the context of random forests, [?, ?, ?, ?] have established consistency results, with [?] showing that under their conditions $\delta_n^2 = \bar{C} \cdot n^{-0.75/(S \log 2 + 0.75)}$, assuming that only S out of the p features play a role in the model. Similarly, [?] and [?] have demonstrated that certain neural networks satisfy Assumption 2; in particular, [?] prove that if $f^{(l)}$ belongs to a Sobolev ball $\mathcal{W}^{\alpha, \infty}([-1, 1]^p)$ with smoothness level α , then one may take $\delta_n^2 = C \cdot (n^{-\alpha/(\alpha+p)} \log^8 n + \log \log n/n)$.

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THEOREM 4.1. Suppose Assumptions 1 and 2 hold. With probability larger than $1 - 1/t^2 - \tau_n$ for $t > 1$ and $\tau_n \rightarrow 0$, the plug-in estimator defined in (3.1) satisfies:

$$\left\| \tilde{f}_{\mathcal{H}} - f_{\mathcal{H}}^* \right\|_{\ell_2(\mathbf{Q})} \lesssim \delta_n + tM \cdot \min \left\{ \delta_n^2 + \frac{M^2}{\sqrt{N}} + \text{Err}_0, \rho_{\mathcal{H}} \right\}, \text{ with } \text{Err}_0 = M\delta_n,$$

where M is the scale of the individual source models as defined in (4.1), δ_n measures the estimation errors of fitted individual models defined in Assumption 2, and $\rho_{\mathcal{H}} := \max_{q, q' \in \mathcal{H}} \|q - q'\|_2$ is the diameter of the set $\mathcal{H}$ that encodes the prior information.

$\min \left\{ \delta_n^2 + \frac{M^2}{\sqrt{N}} + \text{Err}_0, \rho_{\mathcal{H}} \right\}$ captures the error in estimating the optimal aggregation weights. The intuition behind taking the minimum is as follows: if the diameter $\rho_{\mathcal{H}}$ of the set $\mathcal{H}$ is relatively large, the error in the aggregation weight mainly comes from the uncertainty in estimating Γ ; however, if $\rho_{\mathcal{H}}$ is smaller than the statistical precision achieved in estimating the weights, then the constraint imposed by the set $\mathcal{H}$ itself limits the error.

4. Theoretical Justification (Cont.)

For clarity, when $\delta_n \ll M$ —that is, when the estimation error of $\hat{f}^{(l)}$ is small relative to the scale of $f^{(l)}$ —and when the constraint set $\mathcal{H}$ is sufficiently large so that $\rho_{\mathcal{H}}$ does not impose any restriction when computing the aggregation weight, Theorem 4.1 simplifies as

$$\left\| \tilde{f}_{\mathcal{H}} - f_{\mathcal{H}}^* \right\|_{\ell_2(\mathbf{Q})} \lesssim \delta_n + M^2 \delta_n + \frac{M^3}{\sqrt{N}}.$$

In cases where M is large (e.g., in high-dimensional settings where $M = \max_{l \in [L]} \|\beta^{(l)}\|_2$ grows with the dimension p), the second term dominates the convergence rate. In the following Theorem 4.2, we show that the proposed bias-corrected estimator can substantially reduce this dominating term, $M^2 \delta_n$.

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ASSUMPTION 3. There exist positive constants $c_1, c_2 > 0$ such that $\omega^{(l)}(x) \in [c_1, c_2]$ for all $l \in [L]$ and $x \in \mathbb{R}^p$. Furthermore, with probability at least $1 - \tau_n$ with $\tau_n \rightarrow 0$, there exists a positive sequence η_ω such that $\max_{l \in [L]} \|\hat{\omega}^{(l)} - \omega^{(l)}\|_{\ell_4(\mathbf{Q})} \leq \eta_\omega$.

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Here, we give concrete examples of η_ω . In the special setting with no covariate shift, we set $\widehat{\omega}^{(l)} \equiv \omega^{(l)}(\cdot) \equiv 1$ so that $\eta_\omega = 0$. In high-dimensional regimes with covariate shift, we further establish that $\eta_\omega \lesssim \sqrt{s_\gamma \log p / (n + N)}$, where $s_\gamma = \max_{l \in [L]} \|\gamma^{(l)}\|_0$ denotes the maximum sparsity for the high-dimensional logistic density ratio model; see Corollary 3 in Section A.4 of Supplements [?]. Moreover, we present a relaxed version of Assumption 3 in Section C.5 of Supplements [?], where instead of assuming the pointwise boundedness of $\omega^{(l)}$, we control the convergence rate of the relative error $\|\widehat{\omega}^{(l)} / \omega^{(\widehat{l})} - 1\|_{\ell_1(\mathbf{O})}$.

Here, we give concrete examples of η_ω . In the special setting with no covariate shift, we set $\widehat{\omega}^{(l)} \equiv \omega^{(l)}(\cdot) \equiv 1$ so that $\eta_\omega = 0$. In high-dimensional regimes with covariate shift, we further establish that $\eta_\omega \lesssim \sqrt{s_\gamma \log p / (n + N)}$, where $s_\gamma = \max_{l \in [L]} \|\gamma^{(l)}\|_0$ denotes the maximum sparsity for the high-dimensional logistic density ratio model; see Corollary 3 in Section A.4 of Supplements [?]. Moreover, we present a relaxed version of Assumption 3 in Section C.5 of Supplements [?], where instead of assuming the pointwise boundedness of $\omega^{(l)}$, we control the convergence rate of the relative error $\|\widehat{\omega}^{(l)} / \omega^{(\widehat{l})} - 1\|_{\ell_1(\mathbf{O})}$.

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THEOREM 4.2. Suppose Assumptions 1, 2, and 3 hold. With probability at least $1 - 1/t^2 - 2\tau_n$ for $t > 1$ and $\tau_n \rightarrow 0$, the estimator $\hat{f}_{\mathcal{H}}$ defined in (3.2) satisfies:

Theoretical Guarantees (Continued)

- where M is the scale of the individual source models defined in (4.1), δ_n and η_ω measure the estimation errors of the fitted individual models and the density ratio estimators defined in Assumptions 2 and 3, respectively, and $\rho_{\mathcal{H}} := \max_{\mathbf{q}, \mathbf{q}' \in \mathcal{H}} \|\mathbf{q} - \mathbf{q}'\|_2$ is the diameter of the set $\mathcal{H}$ that encodes the prior information.
- In comparison to Theorem 4.1, the bound here replaces the term $\text{Err}_0 = M\delta_n$ in (4.2) with $\text{Err}_1 + \text{Err}_2$.
- The term Err_1 represents the bias estimation error with a known density ratio, while Err_2 captures the bias estimation error due to the density ratio estimation.
- Generally, $\text{Err}_1 + \text{Err}_2$ is smaller than Err_0 , indicating the effectiveness of bias correction.

Simplified Rate for Bias-Corrected Estimator

- Next, we simplify this theorem and discuss how the bias-corrected $\hat{f}_{\mathcal{H}}$ achieves a smaller estimation error compared to the plug-in estimator.
- We consider the scenario where $\delta_n \ll M$ (i.e., the estimation error of $\hat{f}^{(l)}$ is small relative to the scale of $f^{(l)}$) and the constraint set $\mathcal{H}$ is sufficiently large so that $\rho_{\mathcal{H}}$ does not impose a restriction on the aggregation weight estimation.
- Then, Theorem 4.2 simplifies to

$$\|\widehat{f}_{\mathcal{H}} - f_{\mathcal{H}}^*\|_{\ell_2(\mathbf{Q})} \lesssim \delta_n + M^2 \delta_n \cdot (r_n + \eta_\omega) + \frac{M^3}{\sqrt{N}}, \text{ with } r_n = \frac{1}{\sqrt{n \wedge N}} + \frac{\delta_n}{M} + \frac{1}{\delta_n \sqrt{n}}.$$

Interpretation of the Simplified Rate

- By comparing this rate to that of the plug-in estimator in (4.3), we observe that the bias-correction step effectively shrinks the term $M^2\delta_n$ by the multiplicative factor $r_n + \eta_\omega$.
- Notably, r_n diminishes to zero whenever $\delta_n/M \rightarrow 0$ and the convergence rate of the individual estimators $\{\hat{f}^{(l)}\}_{l \in [L]}$ is slower than $n^{-1/2}$, which holds for most ML estimators, including high-dimensional regression, random forests, and deep neural networks.
- Furthermore, if the density ratio estimators are consistent with $\eta_\omega \rightarrow 0$, the overall multiplicative factor $r_n + \eta_\omega$ vanishes, leading to a strictly improved convergence rate.
- In the special case of no covariate shift, where there is no need to estimate density ratios (i.e., $\eta_\omega = 0$), the bias-corrected estimator $\hat{f}_{\mathcal{H}}$ improves precisely by the factor r_n .

Robustness and Summary

- Notably, even if the density ratio estimators are inconsistent, with η_ω remaining at a constant level, the factor $r_n + \eta_\omega$ is still controlled, ensuring that the bias-corrected estimator $\hat{f}_{\mathcal{H}}$ is at least as good as the plug-in estimator.
- This finding highlights the robustness of our proposed bias-correction approach with respect to potential mis-specification in the density ratio model.
- In summary, our analysis demonstrates that the bias-correction step yields an estimator with a smaller estimation error, and it attains a strictly better rate when the density ratio estimators are consistent.
- We also illustrate the numerical advantages of the bias-corrected estimator in Section 5.3, where it consistently outperforms the naive plug-in estimator.

Reward Convergence

- In the discussion above, we quantified the estimation error of the DRoL estimator $\hat{f}_{\mathcal{H}}$ in terms of its $\ell_2(\mathbf{Q})$ distance to the actual $f_{\mathcal{H}}^*$.
- We now translate this result to the objective function by quantifying the difference in rewards.
- **THEOREM 4.3.** For the target distribution $\mathbf{Q}$ with the conditional outcome model defined as $f^{\mathbf{Q}}(X^{\mathbf{Q}}) := \mathbb{E}[Y^{\mathbf{Q}} \mid X^{\mathbf{Q}}]$, we have

$$\left| \mathbf{R}_{\mathbf{Q}}(\hat{f}_{\mathcal{H}}) - \mathbf{R}_{\mathbf{Q}}(f_{\mathcal{H}}^*) \right| \leq 2 \|f^{\mathbf{Q}} - f_{\mathcal{H}}^*\|_{\ell_2(\mathbf{Q})} \cdot \|\hat{f}_{\mathcal{H}} - f_{\mathcal{H}}^*\|_{\ell_2(\mathbf{Q})} + \|\hat{f}_{\mathcal{H}} - f_{\mathcal{H}}^*\|_{\ell_2(\mathbf{Q})}^2,$$

- where the reward function $\mathbf{R}_0(f_{\mathcal{H}}^*)$ is defined in (2.2).

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- This theorem shows that the convergence rate of the reward difference $|\mathbf{R}_Q(\hat{f}_{\mathcal{H}}) - \mathbf{R}_Q(f_{\mathcal{H}}^*)|$ is directly determined by the convergence rate of $\|\hat{f}_{\mathcal{H}} - f_{\mathcal{H}}^*\|_{\ell_2(\mathbf{Q})}$.
- In particular, if $\hat{f}_{\mathcal{H}}$ is a consistent estimator of $f_{\mathcal{H}}^*$ and the distance $\|f^{\mathbf{Q}} - f_{\mathcal{H}}^*\|_{\ell_2(\mathbf{Q})}$ is bounded by a constant, then the reward of our bias-corrected estimator converges to the reward of the population-level model $f_{\mathcal{H}}^*$.
- It is worth noting that Theorem 4.3 also applies to the plug-in DRoL estimator (with $\hat{f}_{\mathcal{H}}$ replaced by $\tilde{f}_{\mathcal{H}}$).
- Theorems 4.1, 4.2, and 4.3 together provide a theoretical framework for analyzing the proposed robust prediction model estimators, addressing the estimation error in both the model parameters and the objective function value.

Application of the Framework

- This framework can be applied to concrete settings by verifying the estimation errors of individual source models and density ratio estimators, as specified in Assumptions 2 and 3.
- In particular, in Section A.4 of Supplements [62], we achieve a rate of $n^{-1/2}$ in the context of low-dimensional linear regression and a rate of $\sqrt{s \log p/n}$ in high-dimensional settings (with s denoting the sparsity level).

- Comparison of Worst-case Reward.
Impact of Prior Information.
Effectiveness of Bias Correction.

- Next, we describe the simulated data setup. The simulation involves generating labeled data from L source domains and unlabeled data from the target domain.
- For each source domain $l \in [L]$, the labeled data $\{X_i^{(l)}, Y_i^{(l)}\}_{i \in [n_l]}$ are i.i.d. generated as follows:

$$X_i^{(l)} \sim \mathcal{N}(0_5, I_5), \quad Y_i^{(l)} = f^{(l)}(X_i) + \varepsilon_i^{(l)}, \quad \text{with} \quad \varepsilon_i^{(l)} \sim \mathcal{N}(0, 1).$$

- We now specify the form of the individual source model $f^{(l)}$ for each $l \in [L]$

$$f^{(l)}(x) = \sin(x^\top \beta^{(l)}) + x^\top A^{(l)}x - \text{Tr}(A^{(l)}), \quad (5.1)$$

- where $\beta^{(l)} \in \mathbb{R}^5$ is generated by drawing each entry independently from $\text{Unif}(-1, 1)$ and $A^{(l)} \in \mathbb{R}^{5 \times 5}$ is generated by drawing each entry independently from $\text{Unif}(-0.5, 0.5)$.

Simulated Data Setup (Continued)

- These parameters $\beta^{(l)}$ and $A^{(l)}$, for $l \in [L]$, are generated randomly once and then fixed throughout the 200 simulation rounds, ensuring consistency across replications.
- For the target domain, the unlabeled data $\{X_j^{\mathbf{Q}}\}_{j \in [N]}$ are i.i.d. drawn from $\mathbf{Q}_X = \mathcal{N}((1, -1, 0.5, 0, 0)^\top, I_5)$.

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- We evaluate our proposed DRoL approach –implemented via Algorithm 1 with the default $\mathcal{H} = \Delta^L$ –against several alternative methods by comparing their worst-case rewards over all distributions in the uncertainty class $\mathcal{C}(\mathbf{Q}_X)$, defined in (2.1).
- Below is a brief description of each method. Except for GroupDRO (which has to be implemented via gradient-based neural networks), all methods, including our DRoL approach, are implemented using XGBoost with 5-fold cross-validation.
- We note that, unlike standard approaches that evaluate performance based on squared error loss, we focus on the reward function; the rationale is detailed in Section 2.3.

Compared Methods

- **ERM.** This method pools all source data and maximizes the empirical reward:

$$\arg \max_{f \in \mathcal{F}} \frac{1}{\sum_{l=1}^L n_l} \sum_{l=1}^L \sum_{i=1}^{n_l} \left[(Y_i^{(l)})^2 - (Y_i^{(l)} - f(X_i^{(l)}))^2 \right].$$

- **ImpWeight.** A common technique in unsupervised domain adaptation is given by reweighting each l -th source data sample by an estimated $\bar{\omega}^{(l)}(\mathbf{x})$ of importance (density) ratio $\omega^{(l)}(\mathbf{x}) = d\mathbf{Q}_X(\mathbf{x})/d\mathbf{P}^{(l)}(\mathbf{x})$. In our implementation, we estimate these density ratios using logistic regression, and then maximize the empirical reward over the pooled, reweighted data:

$$\arg \max_{f \in \mathcal{F}} \frac{1}{\sum_{l=1}^L \sum_{i=1}^{n_l} \hat{\omega}^{(l)}(X_i^{(l)})} \sum_{l=1}^L \sum_{i=1}^{n_l} \left\{ \hat{\omega}^{(l)}(X_i^{(l)}) \left[(Y_i^{(l)})^2 - (Y_i^{(l)} - f(X_i^{(l)}))^2 \right] \right\}.$$

Table 1

- **GroupDRO.** This approach maximizes the worst-case (group-wise) empirical reward across the source domains:

$$\arg \max_{f \in \mathcal{F}} \min_{1 \leq l \leq L} \frac{1}{n_l} \sum_{i=1}^{n_l} \left[(Y_i^{(l)})^2 - (Y_i^{(l)} - f(X_i^{(l)}))^2 \right],$$

implemented via the nested gradient-based algorithm for Group DRO, as detailed in Algorithm 1 of the work [47].

1. *Journal of the American Medical Association*, 2000; 284: 2689-2695.

- We consider scenarios with a varying number of source domains, where L ranges in $\{3, 4, \dots, 10\}$.
- In all experiments, the sample size for the unlabeled target data is fixed at $N = 20,000$.
- In contrast, the source domain sample sizes follow two schemes:
 - ① **Even Mixture:** $n_l = 1000$ for all $l \in [L]$;
 - ② **Uneven Mixture:** $n_1 = 500 \cdot L$ for the first source, and $n_l = 500$ for all $l \geq 2$.
- The results based on 200 simulation rounds are presented in Figure 4.
- Recall that a higher reward indicates better predictive performance.
- Our proposed DRoL method consistently achieves better worst-case reward than the other three methods across all values of L .

Figure 4: Worst-case Reward Comparison

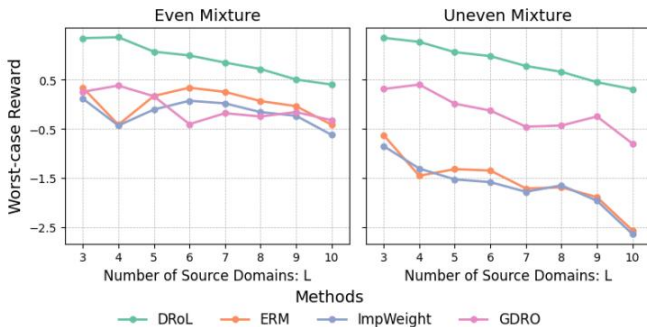


图 4: Comparison of worst-case reward for different methods under even and uneven mixture settings.

1. (a) $\frac{1}{2}$ (b) $\frac{1}{2}$ (c) $\frac{1}{2}$ (d) $\frac{1}{2}$ (e) $\frac{1}{2}$ (f) $\frac{1}{2}$ (g) $\frac{1}{2}$ (h) $\frac{1}{2}$ (i) $\frac{1}{2}$ (j) $\frac{1}{2}$ (k) $\frac{1}{2}$ (l) $\frac{1}{2}$ (m) $\frac{1}{2}$ (n) $\frac{1}{2}$ (o) $\frac{1}{2}$ (p) $\frac{1}{2}$ (q) $\frac{1}{2}$ (r) $\frac{1}{2}$ (s) $\frac{1}{2}$ (t) $\frac{1}{2}$ (u) $\frac{1}{2}$ (v) $\frac{1}{2}$ (w) $\frac{1}{2}$ (x) $\frac{1}{2}$ (y) $\frac{1}{2}$ (z) $\frac{1}{2}$

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- Comparing the left and right panels of Figure 4, we observe that the prediction models produced by ERM and ImpWeight vary markedly between the even and uneven mixture schemes.
- This suggests that their performance is highly sensitive to the relative proportions of each source's sample size when the data are pooled.
- Moreover, even though ImpWeight leverages unlabeled target data, unlike ERM, their worst-case rewards over the uncertainty class $\mathcal{C}(\mathbf{Q}_X)$ are quite similar.
- In contrast, both GroupDRO and our DRoL produce robust prediction models that perform consistently under both the even and uneven mixture schemes.
- Importantly, since GroupDRO is designed for a different setting where target data are not available, and as a result, it does not incorporate unlabeled target data into its formulation.
- Consequently, its robust prediction model is less tailored to the target domain compared to that of DRoL, which leverages the

- Comparison of Worst-case Reward.
Impact of Prior Information.
Effectiveness of Bias Correction.

Impact of Prior Information

- We investigate how different specifications of the constraint set $\mathcal{H}$ affect the predictive performance of DRoL.
- In practice, users may determine the constraint based on their target data $\{X_j^{\mathbf{Q}}\}_{j \in [N]}$.
- In this subsection, we assume a small set of labeled target samples $\{X_k^{\mathbf{Q}}, Y_k^{\mathbf{Q}}\}_{k \in N_0}$ (with $N_0 \ll N$) is available.
- We describe how the labeled target data is utilized in our experiment.
 - ① For each source domain, fit the individual source model $\hat{f}^{(l)}$.
 - ② Determine the optimal aggregation of these fitted individual models that fits the limited labeled target data.
- The optimal weight vector, denoted by $\hat{q}^{\text{label}}$, is obtained via:

$$\hat{q}^{\text{label}} = \arg \min_{q \in \Delta^L} \frac{1}{N_0} \sum_{k=1}^{N_0} \left[Y_k^{\mathbf{Q}} - \sum_{l=1}^L q_l \cdot \hat{f}^{(l)}(X_k^{\mathbf{Q}}) \right]^2. \quad (5.2)$$

1. 0

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- ⚡ We then run Algorithm 1 using this constraint set and refer to the

- We then run Algorithm 1 using this constraint set and refer to the

- resulting method as
- DReL Label**

- resulting method as
- DRoL-Label**
- .

- In the absence of specific domain expertise this procedure

- ✓ In the absence of specific domain expertise, this procedure

- represents one practical way to derive the prior information $\mathcal{H}$

- represents one practical way to derive the prior information μ

- from a limited amount of labeled target data

- from a limited amount of labeled target data.

- More formal approaches that leverage small amounts of labeled

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- 9
- 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 7

- target data with theoretical guarantees are provided in [?].

Experimental Setup for Prior Information

- Assume the target conditional mean satisfies:

$$\mathbb{E}_{\mathbf{Q}}[Y \mid X] = \sum_{l=1}^L q_l^{\mathbf{Q}} \cdot \bar{f}^{(l)}(X),$$

where $q^{\mathbf{Q}} \in \Delta^L$ is the (unknown) true mixture weight vector with $q^{\mathbf{Q}} = (0.6, \frac{0.4}{3}, \frac{0.4}{3}, \frac{0.4}{3})^\top$.

- The small set of labeled target samples $\{X_k^Q, Y_k^Q\}_{k \in N_0}$ are generated as follows:

$$X_k^{\mathbf{Q}} \sim \mathbf{Q}_X, \quad Y_k^{\mathbf{Q}} = \sum_{l=1}^L \gamma_l^{\mathbf{Q}} \cdot f^{(l)}(X_k^{\mathbf{Q}}) + \varepsilon_k^{\mathbf{Q}}, \quad \text{with } \varepsilon_k^{\mathbf{Q}} \sim \mathcal{N}(0, 1),$$

where $\varepsilon_k^{\mathbf{Q}}$ is independent of $X_k^{\mathbf{Q}}$ and the individual source models $\{f^{(l)}\}_{l \in [L]}$ are specified in (5.1).

- We vary $N_0 \in \{20, 50, 100\}$ and $\rho \in [0, 0.9]$.
- Fix $L = 4$, $n_l = 1000$, and $N = 20,000$.

1. *Journal of Management Studies*, 1997, 34, 1, 1-14.

- **DRoL-Unif:** Constructs the constraint set around the uniform weight vector $q^{\text{unif}} = (0.25, 0.25, 0.25, 0.25)^\top$, i.e., $\mathcal{H} = \{q \in \Delta^L \mid \|q - q^{\text{unif}}\|_2 \leq \rho\}$.
- **DRoL-Default:** Without prior information, the proposed DRoL sets $\mathcal{H} = \Delta^L$ as a default.
- **TargetOnly:** Fits a prediction model using only the limited labeled target data, without incorporating any source domain data.

Methods	$N_0 = 20$	$N_0 = 50$	$N_0 = 100$
DRoL-Label	3.0	3.0	3.0
DRoL-Unif	1.5	1.5	1.5
DRoL-Default	1.5	1.5	1.5
TargetOnly	0.0	1.5	2.0

表 2: Example reward values for different methods and N_0 .

Results: Impact of Prior Information (Figure 5)

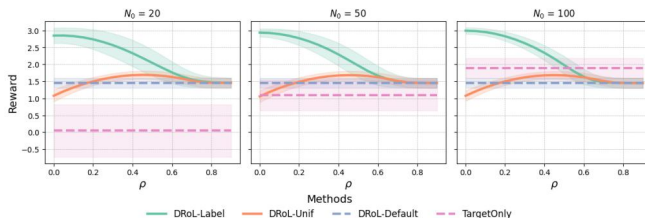


图 5: Comparison of different methods in terms of the reward evaluated on the target distribution $\mathbf{Q}$. The plotted curves represent the mean reward computed over 200 simulation rounds, while the shaded error bands indicate the 10th and 90th percentile variability across these rounds. Here, the target conditional outcome model $\mathbf{Q}_{Y|X}$ is generated as a mixture of the source conditional outcome models $\{\mathbf{P}_{Y|X}^{(l)}\}_{l \in [4]}$ with mixture weights $\gamma^{\mathbf{Q}} = (0.6, \frac{0.4}{3}, \frac{0.4}{3}, \frac{0.4}{3})^\top$. The experiment fixes the number of source domains at $L = 4$ (with $n_l = 2000$ samples), the unlabeled target sample size at $N = 20,000$, and varies $N_0 \in \{20, 50, 100\}$ (which are used only in DRoL-Label and TargetOnly) and the parameter $\rho \in [0, 0.9]$ that controls the size of the constraint set $\mathcal{H}$.

Analysis of Prior Information Results

- Figure 5 presents the results of 200 simulation rounds.
- **DRoL-Label vs. DRoL-Default:** Even a small amount of labeled target data provides valuable insights, enabling DRoL-Label to consistently outperform DRoL-Default across all N_0 and ρ .
- As ρ increases and $\mathcal{H}$ expands to Δ^L , the performance of DRoL-Label converges to that of DRoL-Default.
- **DRoL-Unif:** Simulates a naive assumption that the target domain is an equally weighted average of the source domains.
 - When ρ is small (tight constraint), this naive prior leads to poorer performance compared to DRoL-Default.
 - As ρ increases, the negative impact diminishes, and performance converges to DRoL-Default.
- This highlights that incorporating a mistakenly specified prior can degrade performance, underscoring the importance of carefully setting $\mathcal{H}$ to balance predictiveness and robustness.

1. *Journal of the American Medical Association*, 1997; 277: 1001-1005.

- **TargetOnly** builds the prediction model solely from the limited labeled target data.
- When labeled target samples are too few (e.g., $N_0 = 20$ or 50), the proposed DRoL methods substantially outperform TargetOnly.
- As N_0 increases to 100 , the performance of TargetOnly starts to outperform DRoL-Default (which does not incorporate prior information).
- Nonetheless, **DRoL-Label**, which effectively combines information from both the source domains and the labeled target data, delivers better performance compared to TargetOnly for our considered range of N_0 .

- Comparison of Worst-case Reward.
Impact of Prior Information.
Effectiveness of Bias Correction.

Effectiveness of Bias Correction

- We assess the impact of bias correction in our proposed DRoL Algorithm 1.
- Compare it with a plug-in approach that simply aggregates the ML-fitted individual source models without bias correction.
- Experimental setup:
 - Fix $L = 3$.
 - Vary source sample sizes $n_l \in \{100, 300, 500, 1000\}$.
 - Keep unlabeled target sample size fixed at $N = 20,000$.
 - Use default constraint set $\mathcal{H} = \Delta^L$.
- Compare three variants of DRoL:
 - **PlugIn**: Aggregates individual source models with weights computed directly from ML-fitted models, without bias correction (as in (3.1)).
 - **Logistic**: Estimates density ratios $\{\hat{\omega}^{(l)}\}_{l \in [L]}$ using logistic regression for bias correction (Algorithm 1).
 - **Oracle**: Uses the true density ratios $\{\omega^{(l)}\}_{l \in [L]}$ for bias correction (Algorithm 1).

Results: Effectiveness of Bias Correction (Figure 6)

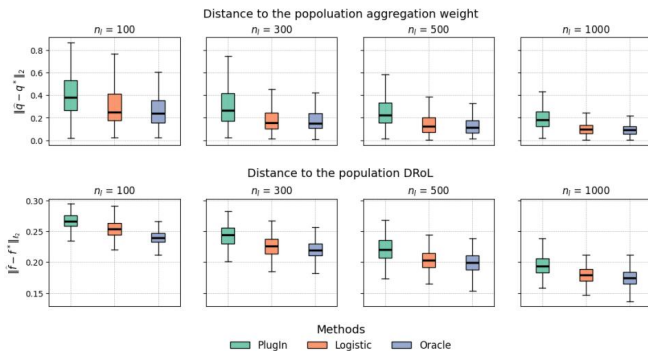


图 6: Comparison of variants of DRoL estimators. PlugIn denotes the plug-in DRoL estimator with $\mathcal{H} = \Delta^L$. In contrast, Logistic and Oracle are bias-corrected DRoL estimators implemented via Algorithm 1 with $\mathcal{H} = \Delta^L$, where density ratios are estimated using logistic regression or set to their true values, respectively. In this experiment, the number of source domains is fixed at $L = 3$, with per-source sample sizes varied over $n_l \in \{100, 300, 500, 1000\}$ and the unlabeled target sample size fixed at

Analysis of Bias Correction Results

- Figure 6 summarizes results over 200 simulation runs.
- **Top panel:** Error $\|\hat{\mathbf{q}} - \mathbf{q}^*\|_2$ (aggregation weight accuracy).
- **Bottom panel:** Error $\|\hat{f} - f^*\|_{\ell_2(\mathbf{Q})}$ (model estimation error).
- **Key findings:**
 - The bias-corrected method **Logistic** outperforms **PlugIn** in both aggregation weight accuracy and model estimation error.
 - Improvements become more pronounced as the source sample size increases.
 - The **Oracle** method, using true density ratios, achieves the best performance.
 - Our **Logistic** method is comparable to the Oracle.
- **Conclusion:** Bias correction is effective and necessary for accurate estimation.

- ## 7 Real Data.

Real Data: Beijing PM2.5 Air Pollution Dataset

- We evaluate DRoL using the Beijing PM2.5 Air Pollution dataset, initially analyzed in [?].
- Contains PM2.5 concentration measurements (2013–2016) from 12 nationally controlled air-quality monitoring sites in Beijing.
- Includes meteorological covariates: temperature, pressure, dew point, precipitation, wind direction, wind speed.
- Motivation:
 - Direct PM2.5 measurement is resource-intensive and requires specialized equipment.
 - Meteorological covariates are widely accessible and exhibit strong associations with PM2.5 levels [?, ?, ?, ?].
 - Useful for predicting PM2.5 with meteorological covariates.

Domain Adaptation Challenge in PM2.5 Prediction

- The relationship between meteorological variables and PM2.5 may vary significantly across districts due to localized environmental factors (e.g., urbanization, industrial activity).
- Models trained solely on data from monitoring sites fail to generalize to districts with distinct meteorological patterns.
- Constructing models that generalize well across Beijing is critical for:
 - Providing early warnings to citizens.
 - Informing regulatory actions to mitigate air pollution.
- This challenge aligns naturally with the Multi-source Unsupervised Domain Adaptation (MSDA) framework.
 - **Labeled source domains:** Air monitoring sites (with paired PM2.5 and meteorological data).
 - **Unlabeled target domains:** Districts where only meteorological measurements are available.
- Goal: Leverage labeled source data and unlabeled target covariates to build robust models that generalize well, considering distribution shifts caused by spatial heterogeneity.

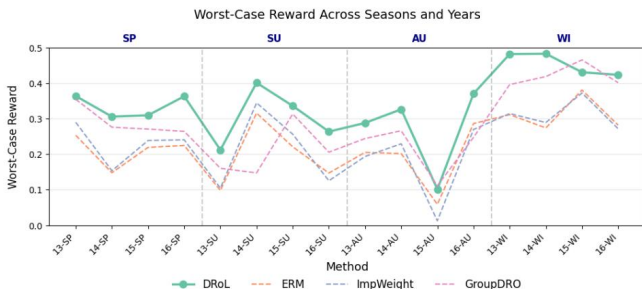
1. *Journal of Management Studies*, 1996, 33, 1, 1-14.

1. $\frac{1}{2}$ 2. $\frac{1}{2}$ 3. $\frac{1}{2}$ 4. $\frac{1}{2}$ 5. $\frac{1}{2}$ 6. $\frac{1}{2}$ 7. $\frac{1}{2}$ 8. $\frac{1}{2}$ 9. $\frac{1}{2}$ 10. $\frac{1}{2}$

1. *Journal of Management Studies*, 1997, 34, 103-117.

- For each year $\in \{2013, \dots, 2016\}$ and for each season (denoted as SP for Spring, SU for Summer, AU for Autumn, and WI for Winter), we develop predictive models based on the five source domains ($L = 5$) and evaluate their performance using the worst-case reward (i.e., the minimum reward) across the seven target sites.
- Consistent with the experiments described in Section 5.1, we compare our proposed DRoL method (implemented via Algorithm 1 with the default constraint $\bar{\mathcal{H}} = \Delta^L$) against ERM, ImpWeight, and GroupDRO that have been described in Section 5.1.

Worst-Case Reward Comparison



1. *Journal of the American Medical Association*, 1997; 277: 1001-1005.

Period	DROL	ERM	ImpWeight	GroupDRO
13-SP	0.37	0.26	0.29	0.35
14-SP	0.31	0.15	0.24	0.28
15-SP	0.31	0.22	0.24	0.27
16-SP	0.37	0.22	0.24	0.27
13-SU	0.21	0.10	0.15	0.15
14-SU	0.40	0.32	0.35	0.15
15-SU	0.34	0.28	0.30	0.31
16-SU	0.27	0.15	0.13	0.21
13-AU	0.29	0.20	0.22	0.23
14-AU	0.33	0.21	0.24	0.27
15-AU	0.10	0.07	0.02	0.15
16-AU	0.38	0.29	0.28	0.30
13-WI	0.48	0.31	0.31	0.40
14-WI	0.48	0.29	0.28	0.45
15-WI	0.43	0.38	0.38	0.47
16-WI	0.42	0.27	0.27	0.41

Worst-Case Reward Comparison (Analysis)

- **FIG 7.** Comparison of methods in terms of the worst-case reward evaluated over seven target domains across seasons and years. The DRoL method is implemented by Algorithm 1 (using default $\mathcal{H} = \Delta^L$). The descriptions of methods ERM, ImpWeight, and GroupDRO are provided in Section 5.1.
- Figure 7 summarizes the results for each year and season. Our proposed DRoL method achieves the highest worst-case reward across all years and seasons, with the only exception of Winter 2015, where its performance is slightly lower than that of GroupDRO.
- While GroupDRO also yields a robust prediction model, the usage of unlabeled target data in the proposed DRoL is tailored to each specific target domain using covariate information, which explains its superior performance.
- Finally, methods like ImpWeight and ERM, which do not account for potential conditional outcome distribution shifts across source and target domains, deliver the poorest

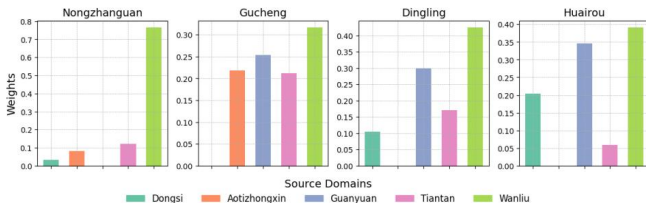
[illegible]

- Moreover, the aggregation weights produced by the proposed DRoL method naturally reflect the relative importance of each source domain for a given target domain.
- For illustrative purposes, Figure 8 displays the aggregation weights computed for four target air-monitoring sites during Spring 2016, enabling a clear visualization of each source air-monitoring site's contribution to building a robust prediction model for each target site.

1. *Journal of the American Medical Association*, 1997; 277: 1001-1005.

- Next, we investigate how different specifications of the constraint $\mathcal{H}$, which encodes prior information about the target distribution, affect the performance of our DRoL approach.
- For DRoL-Label, we consider that 5% of the target domain data is labeled (about 100 labeled samples in this real application), while the remaining 95% of the target domain data is unlabeled. We set $\mathcal{H} = \{\bar{q} \in \Delta^L \mid \|\bar{q} - \hat{q}^{\text{label}}\|_2 \leq \rho\}$, where $\hat{q}^{\text{label}}$ is the aggregation weight estimated using the limited labeled target data (as specified in (5.2)) and $\rho \in [0, 1]$ controls the size of $\mathcal{H}$.
- For DRoL-Unif, we use a uniform prior by setting $\mathcal{H} = \{\bar{q} \in \Delta^{\bar{L}} \mid \|\bar{q} - q^{\text{unif}}\|_2 \leq \rho\}$, with $q^{\text{unif}} = (0.2, 0.2, 0.2, 0.2, 0.2, 0.2)^\top$.
- In contrast, DRoL-Default adopts no prior information by setting $\mathcal{H} = \Delta^L$.
- Additionally, we include the TargetOnly benchmark, which fits models solely using the small amount of labeled target data.

Aggregation Weights Visualization

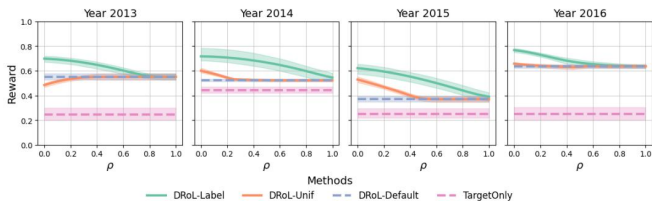


[illegible]

Source Domains	Dongsi	Aotizhongxin	Guanyuan	Tiantan	Wangjiazui
Nongzhanguan	0.03	0.09	0.00	0.12	0.00
Gucheng	0.00	0.23	0.26	0.22	0.00
Dingling	0.11	0.00	0.30	0.17	0.00
Huairou	0.21	0.00	0.36	0.06	0.00

- **FIG 8.** Aggregation weights for four target air-monitoring sites (Nongzhuanguan, Gucheng, Dingling, Huairou) in Spring 2016. The weights are computed by Algorithm 1 using the default constraint set $\mathcal{H} = \Delta^L$.

Reward on Site Nongzhanguan in Summers



Reward on Site Nongzhanguan in Summers (Analysis)

- **FIG 9.** Comparison of different methods in terms of the reward evaluated on the target air-monitoring site Nongzhanguan during the summer season. In each simulation, the models are trained using a randomly selected 90% subsample of the original dataset. The plotted curves are the mean rewards computed over 200 subsampled data sets, and the shaded error bands represent the 10th and 90th percentiles of the rewards across 200 subsampled data sets.
- DRoL-Label, DRoL-Unif, DRoL-Default are implemented via Algorithm 1 with different specifications of the constraint set $\mathcal{H}$.
- DRoL-Label sets $\mathcal{H} = \{q \in \Delta^L \mid \|q - \hat{q}^{\text{label}}\|_2 \leq \rho\}$, where $\hat{q}^{\text{label}}$ is the aggregation weight fitted from a small set of labeled target data (5% of the target dataset), as described in (5.2).
- DRoL-Unif sets $\mathcal{H} = \{q \in \Delta^L \mid \|q - q^{\text{unif}}\|_2 \leq \rho\}$ with a uniform weight vector q^{unif} .
- DRoL-Default uses the default $\mathcal{H} = \Delta^L$, i.e., no prior information.

1. *Journal of Management Studies*, 1997, 34, 1, 1-14.

- Figure 9 illustrates the comparison using data from the target air-monitoring site Nongzhanguan during the summer season.
- When comparing DRoL-Label (which leverages prior information estimated from the limited labeled target data) with DRoL-Default (which does not use any prior information), it is clear that accurately specified prior information can enhance the predictive performance of DRoL.
- Moreover, DRoL-Unif constrains the solution around the uniform vector $q^{\text{unif}} = (0.2, 0.2, 0.2, 0.2, 0.2, 0.2)^T$, failing to capture the target domain's characteristics. As a result, it may perform better or worse than DRoL-Unif in different years.
- Finally, the TargetOnly benchmark, which fits a model solely on the limited labeled target data, performs the worst among these methods and fails to deliver a reliable prediction for the target domain.
- These findings highlight DRoL's ability to integrate prior information, which can improve its predictive performance on a

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- [illegible]

